

TECHNICAL SPECIFICATION (TOR)

1. General Information

Client	inDrive / inVision U
Hackathon	Decentrathon 5.0
Track	AI inDrive
Task Title	Intelligent Candidate Selection Support System for inVision U
Task Type	AI/ML development, Data Science, NLP

2. Problem Description

InVision U is an innovative university with 100% scholarships funded by inDrive, launched in Kazakhstan to educate future leaders, entrepreneurs, and project builders. For the program, not only formal achievements matter, but also motivation, leadership potential, values, and the ability to grow.

Today, the inVision U admissions committee manually evaluates candidate applications, essays, and video presentations. This creates the following problems:

- Talented applicants with a low ability to “sell themselves” are lost before they even come into the selectors’ field of view - the current admissions process sees the application, not the person.
- Standard application forms and essays fail to capture early signals of leadership potential: by the time an application is submitted, many strong candidates have already filtered themselves out.
- Generative AI blurs the authentic voice of the applicant in essays, making traditional text formats an increasingly unreliable assessment tool.
- As the number of applicants grows, the quality of manual screening inevitably declines - scale and depth of assessment are poorly compatible.

3. Stakeholders

Key groups involved in the admissions process:

- inVision U admissions committee
- University administration (involved only at the enrollment stage)
- Applicants (primarily high school students and young adults)
- Local educational partners and government authorities

Key stakeholder pain points:

- How can leadership potential be identified at an early stage - already at school age?
- How can we avoid losing talents who simply described themselves less effectively in the application, but have real potential?
- How can bias be eliminated when selecting “true agents of change”?

4. Hackathon Task

Create an AI/data-driven solution for the initial screening of candidates for inVision U.

The solution must:

- Analyze candidate applications, texts, and interviews.
- Evaluate candidates based on skills, experience, motivation, and potential.
- Generate a recommendation, score, or ranking of the best candidates.
- Significantly reduce the manual workload of the admissions committee.
- Be a transparent support tool for the committee, rather than a fully autonomous decision-making system.

5. Expected Result

Participants need to develop a working prototype in one or a combination of the following formats:

- A web interface or dashboard for the admissions committee.
- A Telegram bot or chat agent for initial interaction with candidates.
- A scoring / ranking engine - an API or script that takes candidate data and returns an evaluation.

Minimum functional requirements for the prototype:

- Accepts input data for one or several candidates.
- Returns a score / recommendation / shortlist with a brief explanation of the key evaluation factors.
- Provides a basic level of explainability (Explainable AI): the committee must understand why a candidate received a particular score.

Required submission artifacts:

- A working prototype (link / demo video / deployment).
- Brief documentation: what the solution does, how to run it, and what data is used.
- A presentation demonstrating the approach and results.

6. Hard Constraints

6.1 Legal and Ethical Boundaries

- Protection of personal data is a priority, especially for minors and young people.

- Algorithmic transparency (Explainable AI) is mandatory: candidates must not be evaluated by a “black box.”
- The solution must not violate applicable personal data protection laws.

6.2 Constraints on the Role of AI

- AI does NOT make final autonomous decisions on admission or rejection. The final word remains with the committee.
- It is unacceptable to use AI without a “human in the loop” for reviewing the context of a candidate profile.

6.3 What Must NOT Be Proposed

- Fully automated “black box” systems for admission/rejection.
- Solutions that score candidates using demographic, racial, or socio-economic data as proxy indicators of success.
- Systems that apply invasive surveillance or social media scraping without the explicit consent of the candidate.

7. Desired Directions

inVision U is particularly interested in the following approaches:

- **Leadership potential identification:** the main goal is to detect real future leaders, not just filter formally strong applications.
- **Assessment of the “path traveled”:** focus on the candidate’s growth trajectory, not only on current achievements.
- **Detection of generative AI usage:** analysis of essays for inauthenticity and possible use of ChatGPT or similar tools.
- **Predictive analytics and “lead scoring”:** tracking signals of candidate engagement and interest before and after application submission.
- **Proactive talent search:** instead of passively waiting for applications, a system that identifies promising candidates through partner networks and educational organizations.
- **Multimodal assessment:** combining structured application data, free-form essay text, bot dialogue data, and other signals.

8. What to Cover in the Presentation

- A description of the proposed solution, its core idea, and its approach to automating the initial screening of candidates.
- How the system analyzes applications and by what parameters it evaluates students, including skills, experience, motivation, and potential.
- How the final recommendation is formed: shortlist, candidate ranking, or selection of the most promising participants.
- What data is used in the solution and how it is processed.
- What practical value the solution provides to the inVision U team, including in terms of reducing manual work and improving selection quality.

- The limitations of the proposed approach, possible risks, and directions for further development of the solution.

9. Evaluation Metrics and Criteria

Criteria	Comments	Points
Problem & Value	How clearly the solution's value for the admissions process and alignment with inVision U's mission are demonstrated.	10
Data & Candidate Representation	Quality of work with sources, features, texts, missing data, and the logic of potential assessment.	15
Baseline & Improvements	Comparison with simple rules / manual scoring / a baseline model is provided, and improvement is shown.	10
Model & Validation	Soundness of the approach, proper metrics, error analysis, and robustness on new profiles.	20
Fairness & Explainability	Transparency of the solution, human-in-the-loop, bias handling, and correctness of result interpretation.	15
Demo & UX	Clear usage scenario, convenient interface / script, strong demonstration of the result.	10
Reliability & Privacy	Anonymization, security, model limitations, and responsible use of data.	10
Documentation	README with run instructions, data description, limitations, dependencies, and usage scenarios.	10
Total		100

10. Deadlines and submission forms

Stage	Period	Stage mechanics
1	Day 1–3 <i>March 27–29</i> 23:59	Development start. You choose the task, shape your solution, and start coding. Submit the following: • GitHub repository with code

		• Solution architecture • A minimally working element (logic / API / module) This stage filters out teams that have not made a real start. Submission form: https://forms.gle/rfTuGW3PE7q74xGY7
2	Day 3–7 <i>March 29 – April 2 23:59</i>	Main development. You implement the solution according to the task requirements. By the end of this stage, you need: • A working prototype • The key task requirements implemented Submit: • GitHub repository • Demo video (how the solution works) • README Submission form: https://forms.gle/JH1Qjnf7VrDHx4h7
	Day 8 <i>April 3</i>	Technical selection. Experts review the projects and choose the teams that move forward.
3	Day 9–10 <i>April 4 – 5 23:59</i>	Final refinement. Teams that advance improve the code, fix errors, and prepare for the final defense. Submit the following: • Final GitHub • Demo video • Presentation Submission form: https://forms.gle/qVZ348mutUJLhQs38 .
	April 5, 23:59	Submission deadline